Bilateral Bargaining with LLM Agents

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1 Introduction

Large language models are increasingly deployed as autonomous agents that interact with environments, tools, and other agents. Recent work has demonstrated that LLM-powered agents can exhibit emergent social behaviours in open-ended simulations [11], coordinate in multi-agent frameworks [19], and interleave reasoning with action [20]. A natural next question is whether these agents behave as economically rational actors when placed in structured bargaining settings where classical theory makes precise, testable predictions. This question has practical significance: automated negotiation is entering consumer and enterprise markets [4], and understanding where LLM bargaining behaviour aligns with—or departs from—established theory is a prerequisite for anticipating the consequences of deployment.

Bilateral bargaining is among the best-characterised settings in economic theory. Rubinstein's alternating-offer model [14] predicts that patient agents extract larger surplus shares through time-preference-based discounting. The anchoring literature [16, 5] establishes that first offers bias final agreements in human negotiation. Roth, Murnighan, and Schoumaker [13] show that human bargainers make disproportionately large concessions near deadlines. At the market level, Rubinstein and Wolinsky [15] predict that decentralised bilateral bargaining produces equilibrium prices but persistent price dispersion—the Law of One Price fails when each pair negotiates independently. Despite the richness of these predictions, existing LLM negotiation studies typically focus on multi-issue games [8, 1], personality-driven dialogue [9], or assistive human–AI interaction [3]. These designs evaluate negotiation capability but do not systematically test LLM behaviour against the specific quantitative predictions listed above. Furthermore, most studies operate at a single level of analysis—either individual bargaining outcomes or aggregate market dynamics—without connecting the two within a single controlled platform.

We address this gap by building a reproducible simulation platform in which LLM agents engage in alternating-offer bilateral bargaining under private economic constraints (value, cost, budget). The platform supports three agent types: a deterministic rule-based baseline that follows a linear concession strategy, an LLM reactive agent that produces single-shot decisions, and an LLM deliberative agent that performs structured reasoning (beliefs, target, strategy, action) before committing to an offer—a design motivated by chain-of-thought prompting [17]. A configurable feasibility gate ensures settlement reliability by accepting clearly profitable offers deterministically, while logging every intervention so that gate-driven and LLM-driven settlements can be distinguished in analysis. Hard constraint enforcement, seeded random number generation, and structured event logging provide the reproducibility infrastructure needed for controlled experimental comparison.

Using this platform, we conduct five experiments totalling 2,250 negotiation sessions. Three experiments—concession dynamics, anchoring bias, and deadline pressure—are replicated across three random seeds (42, 123, 456) for a total of 1,350 sessions. Two further experiments—market price discovery and shock response—are conducted with a single seed (42) for a combined 900 sessions and are reported as exploratory.

This thesis makes three contributions:

1. **A controlled experimental platform** for LLM bilateral bargaining that enforces hard economic constraints, provides deterministic settlement fallbacks, and logs every agent action at sufficient granularity to separate LLM-generated decisions from gate-enforced settlements. The platform connects micro-level bargaining behaviour to macro-level market outcomes within a single system.

2. **Empirical evidence that a 3B-parameter language model** reproduces several qualitative predictions of classical bargaining and market theory—including zone-of-possible-agreement-determined deal success (2.7% deal rate at ZOPA = \$20 versus 100% at ZOPA $\geq$ \$40), market price stabilisation (price trend < \$0.06/tick over 30 ticks in a single-seed run), and lower mean price and liquidity under supply/demand shocks ($-$\$2.15 mean price, -3.1 percentage points liquidity)—while failing to exhibit anchoring bias (first-offer/final-price $r = -0.15$) or deadline-driven concession acceleration. Structured chain-of-thought prompting produces more conservative counter-offers than single-shot prompting (\$7 difference on the first counter), though this is a prompting effect rather than evidence of Rubinstein-style time-preference patience. An important caveat applies: the feasibility gate drives approximately 40% of all acceptances in these experiments, and its interaction with the reported effects is discussed throughout.

3. **An analysis of where and why LLM agents depart from human bargaining heuristics.** The absence of anchoring bias may reflect the agent's prompt-level access to explicit constraint information, though this hypothesis has not been tested via ablation. The absence of deadline sensitivity is confounded by the feasibility gate (which settles deals before deadline proximity becomes relevant) and by prompt instructions that discourage rejection. These null results are reported as characterisations

of LLM bounded rationality under the specific experimental conditions, not as general claims about LLM negotiation capability.

All experiments use a single model (`llama3.2:3b` via Ollama at temperature 0.2). We do not claim that these results generalise across model families or scales; characterising model-specific behaviour under controlled conditions is itself a contribution, and we identify multi-model comparison as an important direction for future work.

The remainder of this thesis is organised as follows. Section 3 reviews related work in LLM-based simulation, LLM negotiation, and classical bargaining theory. Section 4 describes the platform architecture, agent types, validation pipeline, feasibility gate, and reproducibility mechanisms. Section 5 details the experimental design for all five experiments. Section 6 presents the results and analysis. Section 7 discusses implications, limitations, and the role of the feasibility gate. Section 8 concludes.

2 Related Work

3 Related Work

This work sits at the intersection of three research threads: LLM-based agent simulation, LLM negotiation, and classical bargaining theory. We review each in turn and identify the gap that motivates our platform.

3.1 LLM-Based Agent Simulation

The demonstration that large language models can drive believable autonomous agents has opened a productive line of simulation research. Park et al. [11] introduced *Generative Agents*, showing that LLM-powered characters in a sandbox environment exhibit emergent social behaviours—forming relationships, spreading information, coordinating activities—without explicit scripting. This established that LLMs can serve as plausible behavioural engines for agent-based simulation, though the focus was on open-ended social dynamics rather than economic decision-making under constraints.

Subsequent systems extended this paradigm to larger populations and more structured settings. S3 [7] simulates social-network dynamics with LLM agents, studying information diffusion and opinion formation. AgentSociety [10] scales LLM-driven generative agents to large populations to study emergent social phenomena including economic behaviours. Williams and Hosseinmardi [18] applied generative agents to epidemic modelling, demonstrating that LLM agents can reproduce stylised public-health dynamics when placed in structured environments. Gao et al. [6] survey the broader landscape of LLM-empowered agent-based modelling, identifying economic simulation as a high-potential but under-explored application area.

In parallel, frameworks for agentic LLM behaviour have matured rapidly. ReAct [20] introduced a reasoning-and-acting paradigm in which models interleave chain-of-thought reasoning with environment actions—a pattern our deliberative agent adapts for structured negotiation reasoning. Multi-agent orchestration frameworks such as AutoGen [19] provide general infrastructure for LLM agent interaction, but target open-ended tool use rather than controlled economic experiments. Our platform differs from these systems by embedding agents in a well-defined bilateral bargaining protocol with hard economic constraints, enabling experiments that can be compared directly against classical theory.

3.2 LLMs in Negotiation and Bargaining

Negotiation is among the earliest testbeds for learned agents. Lewis et al. [8] introduced end-to-end trained neural negotiation agents using reinforcement learning and self-play on a multi-issue bargaining task, demonstrating that learned models can reach agreements but also degenerate into non-human-like strategies. Their work predates LLM-based approaches and uses task-specific training; our agents are prompted, not trained, and negotiate a single price under private constraints rather than trading across multiple issues.

With the advent of large language models, several works have studied LLM behaviour in negotiation settings. Abdelnabi et al. [1] evaluate LLMs in interactive multi-agent negotiation games, finding that models can reach Pareto-efficient outcomes and that deliberation-style prompting improves agreement quality. Their focus on multi-issue, multi-party games complements our single-issue bilateral design, which isolates price dynamics from issue-trading effects. The finding that structured reasoning aids negotiation quality connects to the chain-of-thought prompting literature [17] and motivates the deliberative agent design in our platform.

Mao et al. [9] study LLM agents with assigned personality traits in multi-issue negotiation games, finding that personality prompts measurably shift outcomes. This motivates our communication-strategy ablation, though we vary prompting strategy rather than personality and measure the effect on bilateral price. Chawla et al. [3] develop assistive LLM agents for socially-aware negotiation dialogues, emphasising cooperative and empathetic communication in human-AI interaction; our agents negotiate autonomously with structured JSON actions, prioritising experimental control over dialogue realism.

Two further works address LLM negotiation from a market perspective. Amato et al. [2] simulate LLM-based autonomous negotiator agents within a game-theoretical framework, studying social behaviour and strategic interaction. Davidson et al. [4] model agent-to-agent negotiations in consumer markets, highlighting both the potential and risks of automated LLM negotiation. Both examine richer market environments than ours; our contribution is the systematic connection between micro-level bargaining behaviour and macro-level market outcomes within a single controlled platform.

3.3 Classical Bargaining and Market Theory

Our experimental design tests several well-established predictions from bargaining and market theory.

Rubinstein's [14] alternating-offer model provides the protocol framework: two agents exchange offers sequentially, with the more patient agent extracting a larger share of surplus. We test whether LLM agents reproduce this patience effect by comparing reactive and deliberative agent types.

The anchoring literature, originating with Tversky and Kahneman [16] and refined in negotiation contexts by Galinsky and Mussweiler [5], predicts that first offers bias final agreements. Our anchoring experiment tests this prediction in LLM agents and finds a null result—a departure from human behaviour that is theoretically informative.

Roth, Murnighan, and Schoumaker [13] established that human bargainers make disproportionately large concessions near deadlines. We test this prediction by varying time horizons with LLM agents.

At the market level, Rubinstein and Wolinsky [15] showed that decentralised bilateral bargaining markets converge to equilibrium prices but exhibit persistent price dispersion, predicting that the Law of One Price fails under bilateral matching. Our market dynamics experiment tests whether LLM markets replicate this structure. Raiffa's [12] concept of the zone of possible agreement (ZOPA) is central to our feasibility gate and to the finding that ZOPA width is the primary determinant of deal success in LLM negotiations.

On the computational side, Zheng et al. [21] demonstrate that multi-agent reinforcement learning can discover effective economic policies in simulated economies, establishing a precedent for using AI agents to study economic mechanisms. Our matching-mechanism experiments are motivated by this line of work, though we use prompted LLM agents rather than RL-trained ones.

3.4 Positioning

Our platform provides a controlled experimental testbed for evaluating LLM bargaining behaviour under economic constraints. It connects three levels of analysis within a single system: (i) micro-level agent behaviour (concession dynamics, anchoring, deadline response), (ii) macro-level market outcomes (price equilibrium, liquidity, surplus distribution, shock response), and (iii) mechanism-design interventions (matching strategies, communication strategies, reputation systems). Existing work typically addresses one of these levels; we span all three using the same agents and protocol, so that the link from individual bargaining behaviour to aggregate market properties can be traced.

The contribution is methodological and empirical rather than algorithmic. We do not propose a new model architecture, a novel learning algorithm, or a fully realistic market simulator. Instead, we provide a reproducible platform with hard safety guarantees—constraint enforcement, feasibility gates, deterministic fallbacks—that enables systematic comparison of LLM behaviour against classical bargaining predictions, and honest reporting of both confirmations and null results.

4 System Design

We describe the simulation platform's architecture, agent types, negotiation protocol, market structure, optional memory and reputation mechanisms, and evaluation infrastructure.

4.1 Architecture Overview

The platform is organised into six layers, each with a single responsibility.

1. **Experiment Control.** CLI entry points and YAML configuration files define experimental conditions: agent types, parameter distributions, matching strategies, seed lists, and sweep grids. A dispatcher routes named experiments to dedicated runner functions that manage replication across seeds and conditions.

2. **Market Simulator.** The tick-level orchestration loop. On each tick the simulator generates buyer and seller populations from configured distributions or fixed parameter lists, applies optional demand/supply shocks (Section 4.6), delegates pair formation to a pluggable `Matcher` (Section 4.6), and dispatches each matched pair to a `NegotiationSession`. In market mode, per-tick statistics (mean price, price dispersion, liquidity) are computed and logged after all sessions in that tick complete.

3. **Negotiation Session.** A stateful object implementing the alternating-offer bargaining protocol [14]. It maintains the transcript, the standing offer, the round counter, and termination state. The buyer always moves first (round 0); agents alternate on subsequent rounds. On each turn, the session constructs an `AgentContext` containing the agent's private constraints, the negotiation history, and the opponent's last offer, then passes it to the active agent's `decide()` method. The session also hosts the feasibility gate (Section 4.5).

4. **Agents.** Each agent implements a `decide(ctx)` → `NegotiationAction` interface. Given context, agents produce a structured action (one of OFFER, COUNTER, ACCEPT, or REJECT) with an associated price, a public message visible to the opponent, and a private rationale logged for analysis but not shown to the opponent. Five agent classes are implemented (Section 4.2).

5. **Action Judge.** A validation and enforcement layer between the agent's raw output and the session's state update. The judge corrects first-round protocol errors, validates offers against hard constraints (budget, cost, price bounds), and replaces invalid actions with REJECT while emitting a structured risk event (Section 4.4).

6. **Settlement and Metrics.** Non-LLM computation of surplus, welfare, and aggregate statistics. When a session terminates with acceptance, buyer surplus (value − price) and seller surplus (price − cost) are computed deterministically. Aggregate metrics are computed over all sessions in a run (Section 4.7).

This separation enables controlled experimentation: agent types, matching strategies, and prompt configurations can be varied independently without modifying the negotiation protocol. Agents never validate their own actions, the session never constructs prompts, and the simulator does not inspect negotiation transcripts.

4.2 Agent Types

All agents share a common interface: given an `AgentContext` containing the item under negotiation, the agent's role and private constraints (value and budget for buyers; cost and target margin for sellers), the current round, the negotiation history, and the opponent's last offer, the agent returns a `NegotiationAction`. Buyers face a dual constraint: the effective price ceiling is $\min(\text{value}, \text{budget})$.

Rule-based baseline. A deterministic agent following a linear concession strategy. Buyers open at 50% of their price ceiling and concede linearly toward it over the available rounds; sellers open at cost $\times$ $(1 + 2m)$, where m is the target margin (default 0.15), and concede toward cost. Each round, the agent computes a concession target as

$$t_r = p_0 + (p_{\max} - p_0) \cdot \frac{r}{R-1}$$

where p_0 is the opening price, $p_{\max}$ is the constraint boundary, r is the current round, and R is the maximum number of rounds. The agent accepts when the opponent's standing offer meets or exceeds this target. On the final round, agents accept any offer within their hard constraints to avoid deadlock. This agent produces zero variance across seeds, providing a deterministic baseline for isolating LLM-specific effects.

LLM reactive. A single-shot LLM agent. Each prompt is assembled from modular components controlled by a `PromptConfig` object:

- A JSON-only guard and system preamble describing the alternating-offer protocol.
- The agent's private constraints, including explicit action rules (e.g. suggested opening price, per-round concession increment, and the instruction to prefer COUNTER over REJECT).
- The objective function stated as an equation (e.g. "maximise buyer_surplus = value − deal_price").

- Round and deadline information, with configurable deadline-salience warnings when few rounds remain.
- An accept-condition block showing the opponent's last offer, the agent's acceptance threshold, and the pre-computed utility, so the model need not perform arithmetic.
- A compressed transcript of the k most recent turns (default k=6), formatted as terse bullet points (e.g. `R0 buyer OFFER $60.00`).
- The required JSON schema with two few-shot examples (one ACCEPT, one COUNTER).

The prompt explicitly instructs the model to prefer COUNTER over REJECT, on the grounds that any deal yields positive surplus while no deal yields zero. This steering reduces walk-away behaviour but does not prevent the model from rejecting if it chooses to.

The LLM's response is parsed, validated, and if malformed, retried once with a format-correction prompt. If the retry also fails, a deterministic fallback action is generated using the same linear-concession logic as the rule-based agent. This shared fallback ensures that LLM failures do not produce undefined behaviour.

LLM deliberative. Identical to the reactive agent except the prompt appends four explicit reasoning steps that the agent must complete inside its `rationale_private` field before committing to an action:

1. BELIEFS—estimate the opponent's reservation price from observed offers.
2. TARGET—compute an ideal price given the remaining rounds.
3. STRATEGY—choose whether to concede, hold firm, or accept.
4. ACTION—specify the concrete action and price.

This design draws on chain-of-thought prompting [17]: by requesting explicit intermediate reasoning, the prompt encourages the model to decompose the negotiation problem before acting. Whether the model performs genuine multi-step reasoning or generates post-hoc rationalisation is not verified; we measure only the observable effect on bargaining outcomes.

Memory agent. Extends the deliberative agent with episodic memory. A `MemoryStore` records the outcome of each completed negotiation (deal status, price, rounds, and an inferred opponent-style label), and the k most relevant past experiences (default k=5) are injected into the prompt as context. Relevance is determined by a simple heuristic: prefer interactions involving the same item, then the most recent. Memory is maintained in-memory for the duration of a single simulation run and is not persisted across runs.

Reputation agent. Further extends the memory agent with a `ReputationStore` that maintains per-opponent interaction histories. For each known opponent, the store computes a summary: number of past interactions, deal rate, average deal price, average negotiation length, and an inferred style label. Style labels are assigned by hard-coded thresholds: *cooperative* if deal rate ≥ 0.8 and average rounds ≤ 3; *moderate* if deal rate ≥ 0.5; *unreliable* if deal rate < 0.3; *tough* otherwise. This reputation summary is prepended to the prompt so the agent can condition its strategy on the opponent's observed track record. The reputation agent requires per-agent memory stores (rather than role-level shared stores) and repeated pairings across ticks, provided by the round-robin matcher (Section 4.6).

4.3 Negotiation Protocol

Each negotiation session implements a bounded alternating-offer protocol. The buyer moves on even-numbered rounds (starting at round 0) and the seller on odd-numbered rounds. This fixed turn order gives the buyer a first-mover advantage, which we report rather than attempt to eliminate.

A session terminates under one of three conditions:

- **Accepted:** an agent outputs ACCEPT, settling at the opponent's last offered price.
- **Rejected:** an agent outputs REJECT, ending the negotiation with no deal.
- **Timeout:** the maximum number of rounds R is reached without agreement.

Upon acceptance, surplus is computed deterministically: buyer_surplus = value − price and seller_surplus = price − cost. Total welfare is the sum of both surpluses, which equals value − cost and is independent of the agreed price.

4.4 Action Validation Pipeline

Every action produced by an agent passes through a three-stage validation pipeline before updating session state.

Stage 1: JSON extraction and schema validation. For LLM agents, the raw text output must be parsed into a structured action. The parser attempts four extraction strategies in sequence: (i) direct JSON parsing, (ii) extraction from markdown code fences, (iii) brace-delimited substring extraction, and (iv) heuristic repair (single-to-double quote conversion, trailing comma removal, unquoted key correction). The extracted object is checked against the action schema: it must contain an `action` field with a value in {OFFER, COUNTER, ACCEPT, REJECT}, an `offer_price` (required and positive for OFFER/COUNTER, coerced to null for ACCEPT/REJECT), and a `message_public` string.

Stage 2: Protocol and constraint validation. The `ActionJudge` applies domain-level checks. First-round corrections handle protocol violations: a COUNTER on round 0 is relabelled as OFFER with the proposed price preserved; an ACCEPT or REJECT on round 0 is replaced with an OFFER at the agent's proposed price if available, otherwise at a default opening price (50% of value for buyers, $1.5 \times$ cost for sellers). Constraint validation checks hard limits: buyer offers must not exceed budget or value; seller offers must not fall below cost; all prices must lie within configurable global bounds $[p_{\min}, p_{\max}]$.

Stage 3: Enforcement. If validation fails, the judge replaces the action with REJECT and emits a structured risk event recording the violation type (budget, cost, bounds, or logic), the attempted action and price, the round number, and the time step. Risk events are logged in the JSONL event stream and aggregated in the run summary, enabling post-hoc analysis of constraint-violation frequency across agent types and conditions.

4.5 Feasibility Gate

A configurable deterministic settlement mechanism that can override agent decisions at two points within each negotiation turn. Both components are controlled by a single boolean flag (`gate_enabled`, default `true`); when disabled, neither fires.

Mechanism. Before calling the agent's `decide()` method, the session evaluates whether the opponent's standing offer is feasible for the current agent. Feasibility requires non-negative utility: for a buyer, the offer must satisfy offer $\leq$ min(value, budget); for a seller, offer $\geq$ cost [?, cf.]]raiffa1982. If feasible, the session settles immediately without invoking the LLM. This addresses a practical failure mode: small language models frequently fail to output ACCEPT even when the offer is clearly profitable, causing deadlock when a zone of possible agreement (ZOPA) unambiguously exists.

A secondary post-LLM check applies the same feasibility test after the agent acts, overriding COUNTER or REJECT to ACCEPT if the standing offer is feasible. Both gate components log the LLM's intended action alongside the effective (possibly overridden) action, an override flag, and the computed utility, allowing post-hoc separation of LLM-generated decisions from gate-driven settlements.

Trade-off. The gate ensures high deal rates in wide-ZOPA scenarios but masks behavioural effects requiring multi-round interaction, notably deadline pressure and extended concession trajectories. Experiments targeting these effects disable the gate and accept lower deal rates as part of the measurement. We report gate-enabled results alongside the gate's override rate so readers can assess whether settlement outcomes are agent-driven or mechanically enforced.

4.6 Market Structure

Population generation. The simulator supports two scenario modes. In *distribution mode*, buyer values, budgets, seller costs, and target margins are drawn independently from uniform distributions with configurable bounds (e.g. buyer value $\in [\$80, \$150]$, seller cost $\in [\$50, \$120]$). In *fixed mode*, all agents receive identical parameter values specified in the configuration, enabling controlled single-variable experiments. In both modes, populations are regenerated each tick from a forked RNG, so agents within a tick are independent draws.

Matching strategies. Pair formation is delegated to a pluggable `Matcher` interface. Four implementations are provided:

- **Random** (default). Shuffles buyers and sellers independently and pairs them 1:1. Unmatched agents (when population sizes differ) are dropped. This is the baseline matching mechanism used in experiments A–E.
- **Surplus-maximising** (oracle). Computes ZOPA = buyer value−seller cost for all possible pairs, then greedily assigns pairs in descending ZOPA order (each agent used at most once). Only pairs with positive ZOPA are matched. This uses private information and represents a theoretical upper bound on matching efficiency, not a realistic mechanism.
- **Sorted** (heuristic). Sorts buyers by value descending and sellers by cost descending, then pairs them positionally. This tends to create moderate but consistent ZOPAs without requiring private information beyond the ordering.
- **Round-robin** (repeated). On the first tick, pairs agents 1:1 and remembers the pairings. On subsequent ticks, agents with the same IDs are re-paired, enabling reputation and memory effects to accumulate across interactions. New agents are paired randomly with unmatched partners.

Demand and supply shocks. An optional shock mechanism applies multiplicative perturbations to buyer values and seller costs at the start of each tick. Shocks fire with a configurable probability (default 10%) per tick. When a shock fires, buyer values are scaled by a demand multiplier drawn uniformly from a configured range (e.g. $[0.7, 1.3]$) and seller costs are scaled by an independently drawn supply multiplier from the same range. Budgets are not modified by shocks, preserving hard spending constraints. This mechanism enables experiments on market resilience and price response to exogenous perturbations.

4.7 Metrics and Evaluation

Session-level metrics. Each completed negotiation produces a structured result containing: deal status, agreed price (if any), number of rounds, termination reason, buyer and seller surplus, and a list of risk events.

Run-level aggregates. After all sessions in a run complete, the following metrics are computed:

- **Deal success rate**: fraction of sessions resulting in a deal.
- **Price statistics**: mean, median, and standard deviation of deal prices.
- **Surplus**: mean buyer surplus, mean seller surplus, and mean total welfare.
- **Rounds**: mean rounds to close (deals only) and mean rounds across all sessions.
- **Deadlock rate**: fraction of sessions terminating by timeout.
- **Risk events**: counts of budget violations, cost violations, and total constraint violations.

Tick-level statistics (market mode). In market mode, the simulator computes per-tick aggregates after each tick's sessions complete: number of sessions, deals made, fail rate, mean transaction price, within-tick price standard deviation, liquidity (deals / sessions), and mean buyer and seller surplus. These are logged as `tick_end` events and enable time-series analysis of market dynamics.

Efficiency and inequality. Two additional metrics support mechanism-comparison experiments. *Allocative efficiency* is the ratio of realised total welfare to the theoretical maximum welfare (computed as $\sum \max(\text{value}_i - \text{cost}_i, 0)$ over all matched pairs, where unrealised deals contribute zero welfare). The *surplus Gini coefficient* measures inequality in surplus distribution across deals, with 0 indicating perfect equality and 1 indicating maximum concentration.

4.8 Reproducibility

Three mechanisms ensure that experiments are reproducible across runs.

Seeded random number generation. All stochastic components—buyer/seller parameter generation, item assignment, pair matching, shock application—draw from a `SeededRNG` wrapper around Python's `random.Random`. At each tick, the simulator calls `fork()`, which consumes one value from the parent RNG to derive a child seed, producing an independent child generator for that tick. This isolates per-tick randomness so that adding or removing ticks does not alter the random sequences of unaffected ticks. LLM inference introduces non-determinism from the model itself, but all market-level processes remain fully reproducible given the same seed.

Configuration-driven experiments. All experimental parameters—agent types, negotiation limits, market distributions, shock probabilities, matching strategies, prompt configurations—are specified in YAML files with CLI override support. Each experiment runner records the active configuration, the git commit hash, and the seed list in a structured JSON summary, enabling exact replication.

Structured event logging. Every negotiation action, session outcome, constraint violation, and tick-level aggregate is written to a newline-delimited JSON log (`events.jsonl`). Turn events include the round number, role, action type, offer price, and public message. For LLM agents, the log additionally records the intended action (what the model output), the effective action (what the simulator executed), an override flag, the computed utility, and the feasibility threshold. This granular logging supports post-hoc analysis—including separation of LLM-driven and gate-driven settlements—without re-execution.

5 Experimental Setup

This section describes the design of all eight experiments conducted on the platform. Five experiments (A–E) are complete, with results reported in Section 6. Three further experiments (F–H) have infrastructure implemented but have not yet been executed; their designs are documented here for completeness.

5.1 Common Setup

All experiments share the following configuration unless explicitly overridden.

Model and inference. A single language model is used throughout: `llama3.2:3b` served locally via Ollama at temperature 0.2, with a maximum output length of 256 tokens and a 30-second timeout per call. Failed LLM calls are retried once with a format-correction prompt; if the retry also fails, a deterministic fallback action is generated using the rule-based linear concession formula (Section 4.2).

Feasibility gate. The deterministic settlement gate (Section 4.5) is enabled for all five completed experiments. As discussed in Section 4.5, this ensures high deal rates in wide-ZOPA scenarios but masks behavioural effects that require multi-round interaction. The gate's override rate is reported per experiment in Section 6 so that gate-driven and LLM-driven settlements can be distinguished.

Prompt steering. Two prompt-level design choices apply across all LLM conditions and affect interpretation:

- The prompt explicitly instructs the model to prefer COUNTER over REJECT, reducing walk-away behaviour. This steering is active in all experiments and may suppress deadline effects (Experiment C) and inflate deal rates relative to an unsteered baseline.

- An accept-condition block pre-computes the utility of the opponent's standing offer and presents it directly in the prompt, so the model need not perform arithmetic. This may reduce anchoring effects (Experiment B) by giving the model explicit surplus information rather than requiring it to infer value from the first offer.

Neither of these design choices has been ablated; their effects on results are discussed as confounds rather than controlled variables.

Reproducibility. All stochastic processes are controlled by a `SeededRNG` with per-tick forking (Section 4.8). Each experiment records the active configuration, seed list, and git commit hash in a structured JSON summary. Every agent action, session outcome, and constraint violation is logged to a JSONL event stream.

Replication. Experiments A–C are each replicated across three random seeds (42, 123, 456). Experiments D and E use a single seed (42) and are reported as exploratory. Table 1 summarises all eight experiments.

Table 1: Summary of all experiments. Status indicates whether results are reported in this thesis.

Exp	Name	Independent variable	Agent type	Sessions	Status
A	Concession	agent type (3 levels)	mixed	450	Complete (3 seeds)
B	Anchoring	ZOPA width (3 levels)	reactive	450	Complete (3 seeds)
C	Deadline	max rounds (3 levels)	deliberative	450	Complete (3 seeds)
D	Market	time (30 ticks)	reactive	450	Complete (1 seed)
E	Shock	shock on/off	reactive	900	Complete (1 seed)
F	Reputation	reputation memory	reputation	—	Infrastructure only
G	Communication	comm. strategy (4 levels)	reactive	—	Infrastructure only
H	Mechanism	matching algorithm (4 levels)	reactive	—	Infrastructure only

5.2 Experiment A: Concession Dynamics

Research questions.

RQ-A1. Do LLM agents exhibit systematic concession patterns across negotiation rounds?

RQ-A2. How do concession trajectories differ between rule-based, reactive, and deliberative agents?

RQ-A3. Does chain-of-thought prompting (deliberative) alter concession behaviour relative to single-shot prompting (reactive)?

Independent variable. Agent type, varied across three conditions: `rule_based`, `llm_reactive`, and `llm_deliberative`. Both buyer and seller use the same agent type within each condition.

Controlled variables. All sessions use fixed-mode parameters to isolate the effect of agent type from scenario variation. The fixed scenario is: buyer value = \$120, seller cost = \$80, buyer budget = \$130, item reference price = \$100, seller target margin = 15%. This creates a ZOPA of \$40 (\$120 − \$80), wide enough for reliable settlement. Maximum rounds = 10. Feasibility gate is enabled.

Design. 3 agent-type conditions × 3 seeds × 50 sessions per seed = 450 sessions total.

Metrics.

- Deal rate (%) per condition.
- Mean agreed price (\$ ± standard deviation across seeds).
- Mean rounds to settlement.
- Per-round offer price trajectory (concession curve), plotted per agent type and role.
- Buyer and seller surplus.
- Gate involvement rate: fraction of acceptances driven by the feasibility gate rather than agent decision.

Hypotheses.

H-A1. All three agent types achieve >90% deal rate given the wide ZOPA.

H-A2. LLM agents settle in fewer rounds than the rule-based agent, whose linear concession formula spreads concessions across the full horizon.

H-A3. The deliberative agent produces different counter-offer prices than the reactive agent, reflecting the effect of structured reasoning on offer generation.

5.3 Experiment B: Anchoring Effects

Research questions.

RQ-B1. Does ZOPA width—manipulated via buyer value—affect settlement price?

RQ-B2. Do LLM agents exhibit anchoring bias, where the first offer pulls the final price toward it?

Independent variable. Buyer value, varied across three levels: \$100, \$120, and \$150. Seller cost is fixed at \$80, giving ZOPA widths of \$20, \$40, and \$70 respectively. Buyer budget is set \$10 above buyer value in each condition (\$110, \$130, \$160) to avoid budget-binding confounds.

Controlled variables. Agent type is `llm_reactive` in all conditions. Seller target margin = 15%, item reference price = \$100, maximum rounds = 10, feasibility gate enabled.

Design. 3 ZOPA conditions × 3 seeds × 50 sessions per seed = 450 sessions total.

Metrics.

- Deal rate per ZOPA level.
- Mean settlement price per ZOPA level (\$ ± standard deviation).
- Mean first offer (buyer's round-0 price) per condition.
- Pearson correlation r between first-offer price and final deal price, computed across all deals.
- Buyer surplus per condition.

Hypotheses.

H-B1. Wider ZOPA leads to higher deal rates. Narrow ZOPA (\$20) may produce deal failure if the seller's opening counter exceeds the buyer's hard cap.

H-B2. If anchoring bias is present, the first-offer/final-price correlation r should be positive and significant: higher first offers should pull settlement prices upward.

H-B2′. If the accept-condition block overrides anchoring, settlement prices may be insensitive to first-offer level, yielding $r \approx 0$.

5.4 Experiment C: Deadline Pressure

Research questions.

RQ-C1. Does varying the negotiation horizon (maximum rounds) affect when deals settle?

RQ-C2. Do LLM agents exhibit deadline effects—accelerated concession near the final rounds—consistent with the findings of Roth et al. [13]?

Independent variable. Maximum rounds per session, varied across three levels: 4, 8, and 16.

Controlled variables. Agent type is `llm_deliberative` in all conditions, chosen because the deliberative agent's structured reasoning offers the best opportunity for deadline-aware strategy adjustment. Fixed scenario parameters match Experiment A (buyer value = $120, seller cost = $80, ZOPA = $40). Feasibility gate is enabled. The prompt includes configurable deadline-salience language that warns the agent when few rounds remain.

Design. 3 horizon conditions × 3 seeds × 50 sessions per seed = 450 sessions total.

Metrics.

- Deal rate per horizon condition.
- Mean settlement round per condition (mean ± standard deviation).
- Settlement round as a fraction of maximum rounds (normalised timing).
- Share of agreements occurring in the last two rounds of the available horizon.

Hypotheses.

H-C1. If agents are deadline-sensitive, longer horizons should produce later settlements (agents use available time) and a disproportionate share of agreements near the deadline.

H-C1′. If the feasibility gate dominates settlement timing, the settlement round should be constant regardless of horizon, as the gate fires whenever the standing offer becomes feasible—typically in the first few rounds.

Known confounds. Two design choices work against observing deadline effects. First, the feasibility gate settles deals as soon as offers become feasible, typically before deadline proximity is relevant. Second, the prompt steers agents to prefer COUNTER over REJECT, removing the primary mechanism by which agents could delay settlement. These confounds are not ablated in this experiment; their implications are discussed in Section 6.

5.5 Experiment D: Market Price Discovery

Research questions.

RQ-D1. Does a multi-tick market populated by LLM bilateral negotiators converge toward a stable price level?

RQ-D2. Does price dispersion persist within ticks, consistent with the bilateral bargaining prediction that the Law of One Price fails when pairs negotiate independently [15]?

Independent variable. Time, observed longitudinally over 30 market ticks. There is no between-condition manipulation; this is a single-condition observational study.

Controlled variables. The simulator runs in market mode with 15 buyer–seller pairs per tick, using random matching. Agent type is `llm_reactive`. Scenario mode is *distribution*: buyer values are drawn uniformly from [$80, $150], seller costs from [$50, $120], buyer budgets from [$100, $200], and seller margins from [5%, 25%]. Maximum rounds = 10. Feasibility gate is enabled.

Design. 30 ticks × 15 sessions per tick = 450 sessions, single seed (42). This is an exploratory design; results should be interpreted with the single-seed caveat.

Metrics.

- Mean transaction price per tick, with within-tick standard deviation.
- Liquidity per tick: fraction of matched pairs that reach a deal.
- Price trend: slope of a simple linear regression of mean price over ticks.
- Mean buyer surplus and mean seller surplus per tick.
- Early-vs-late comparison: mean price and liquidity in ticks 0–4 versus ticks 25–29.

Hypotheses.

H-D1. Mean price stabilises within the first 10 ticks, with a near-zero price trend slope thereafter.

H-D2. Within-tick price dispersion remains substantial (standard deviation > $10), consistent with bilateral bargaining theory.

H-D3. Buyer surplus exceeds seller surplus on average, reflecting the buyer's first-mover advantage in the alternating-offer protocol.

5.6 Experiment E: Shock Response

Research questions.

RQ-E1. Do exogenous demand and supply shocks produce measurable shifts in market price and liquidity?

RQ-E2. Is the direction of price adjustment consistent with economic theory (demand increase → price increase; supply increase → price decrease)?

Independent variable. Shock condition, varied across two levels: `no_shock` (baseline) and `with_shock`. In the shock condition, each tick has a 30% probability of a shock event. When a shock fires, buyer values are scaled by a demand multiplier drawn uniformly from [0.7, 1.3] and seller costs are scaled by an independently drawn supply multiplier from the same range. Budgets are not modified.

Controlled variables. Both conditions use the same market-mode configuration as Experiment D: 30 ticks, 15 pairs per tick, random matching, distribution-mode parameters, `llm_reactive` agents, feasibility gate enabled.

Design. 2 conditions $\times$ 30 ticks $\times$ 15 sessions per tick = 900 sessions total, single seed (42). Exploratory design.

Metrics.

- Mean transaction price per tick, compared across conditions.
- Within-tick price standard deviation per condition.
- Mean liquidity per condition, with minimum and maximum tick-level values.
- Price and liquidity differences (Δ) between shock and no-shock conditions.

Hypotheses.

H-E1. The shock condition produces higher price volatility (larger tick-to-tick price variance) than the baseline.

H-E2. Mean price and mean liquidity are lower under shocks, because negative shocks (demand multiplier < 1 or supply multiplier > 1) compress the ZOPA for affected pairs.

H-E3. Worst-case liquidity drops more severely under shocks than best-case liquidity improves, producing an asymmetric tail risk.

5.7 Planned Extensions

Three additional experiment families have full infrastructure implemented in the platform but have not been executed. Their designs are documented here to support future work and to clarify why the corresponding platform components (reputation store, communication strategies, alternative matchers) exist in the codebase.

5.7.1 Experiment F: Reputation and Repeated Interaction

Research question. Does access to opponent reputation information improve deal rates and surplus in a repeated-interaction market?

Design overview. The round-robin matcher (Section 4.6) pairs agents repeatedly across ticks, enabling the reputation agent (Section 4.2) to accumulate interaction histories. The reputation store tracks per-opponent deal rate, average price, average negotiation length, and an inferred style label (cooperative, moderate, tough, or unreliable). This summary is injected into the agent's prompt so it can condition strategy on the opponent's observed behaviour.

Independent variable. Agent type: `llm_deliberative` (no reputation) versus `reputation` (with reputation store).

Metrics. Deal rate, mean price, surplus, and negotiation length, compared between conditions over 30+ ticks. Of particular interest is whether deal rates improve over time as reputation information accumulates.

Note. The feasibility gate must be disabled for this experiment, as it would mask the strategic adaptation that reputation information is intended to produce.

5.7.2 Experiment G: Communication Strategy

Research question. Does the framing and tone of negotiation messages—controlled via prompt-level communication strategies—affect concession speed and settlement price?

Design overview. The platform supports four communication strategies, each implemented as a prompt modifier in the `PromptConfig`: *neutral* (default factual tone), *assertive* (firm, confident language), *collaborative* (emphasising mutual benefit), and *strategic* (explicit surplus-maximisation framing). These strategies modify the stylistic guidance in the prompt but do not alter the agent's private constraints or the negotiation protocol.

Independent variable. Communication strategy $\in$ {neutral, assertive, collaborative, strategic}, applied to both agents in each session.

Metrics. Deal rate, mean settlement price, mean rounds to settlement, and surplus split across the four conditions.

Note. The feasibility gate should be disabled for this experiment to allow communication-driven effects to manifest over multiple rounds.

5.7.3 Experiment H: Market Mechanism Design

Research question. Does the matching algorithm affect market-level welfare, fairness, and efficiency?

Design overview. The platform provides four matching algorithms (Section 4.6): random (baseline), surplus-maximising (oracle upper bound), sorted (heuristic), and round-robin (repeated interaction). Each matcher determines which buyer–seller pairs are formed, affecting the distribution of ZOPAs and therefore the feasibility and surplus of negotiations.

Independent variable. Matching algorithm $\in$ {random, surplus-max, sorted, round-robin}, with all other market parameters held constant.

Metrics. Allocative efficiency, surplus Gini coefficient, mean deal rate, and mean transaction price across matchers (Section 4.7). The surplus-maximising matcher provides a theoretical upper bound against which the other mechanisms can be evaluated.

6 Results

This section reports results for the five completed experiments (A–E) described in Section 5. We begin with infrastructure-level observations that apply across all experiments, then present per-experiment findings, and conclude with a cross-experiment synthesis.

6.1 LLM Quality and Gate Summary

Before examining experimental outcomes, we report the reliability of the LLM backend and the feasibility gate's involvement across all 2,250 sessions.

Table 2: LLM backend quality and feasibility gate involvement across all experiments.

Metric	Value
LLM fallback rate (format failures)	0.0%
Post-LLM override rate	0.0%
Gate-driven acceptances (fraction of all turns)	~40%
LLM-driven decisions (fraction of all turns)	~60%

The `llama3.2:3b` model produced valid JSON on every call across all experiments, requiring zero fallback actions. The post-LLM feasibility check (which overrides COUNTER or REJECT to ACCEPT when the standing offer is feasible) was never triggered, indicating that when the gate did not pre-empt the LLM call, the model never rejected a clearly profitable offer.

The gate drove approximately 40% of all acceptances. In fixed-scenario experiments (A–C), the gate fires deterministically after the seller's round-1 counter-offer whenever that counter is within the buyer's constraints, settling all sessions at round 2. In market-mode experiments (D–E), gate involvement varies per session depending on the randomly drawn ZOPA for each pair.

Experiment E produced 2 JSON parse failures and 1 LLM timeout across 900 sessions; the retry and fallback mechanisms handled all three without data loss.

6.2 Experiment A: Concession Dynamics

Table 3 reports the mean offer price at each observed round for the three agent-type conditions. All 450 sessions (150 per condition) resulted in deals, confirming H-A1.

Table 3: Mean offer price by agent type and round ($n = 150$ per condition, pooled across 3 seeds). Concession Δ is the absolute difference between the seller's round-1 counter and the buyer's round-0 opening.

Condition	Round 0 (Buyer opens)	Round 1 (Seller counters)	Concession Δ
`rule_based`	\$60.00 ± \$0.00	\$101.33 ± \$0.00	\$41.33
`llm_reactive`	\$65.46 ± \$0.90	\$103.78 ± \$3.05	\$38.32
`llm_deliberative`	\$66.74 ± \$0.67	\$96.57 ± \$2.72	\$29.83

All sessions settled at round 2 via the feasibility gate, which fired after the seller's round-1 counter-offer. This limits the concession data to two observations per session.

Finding 1: LLM agents open less aggressively than the rule-based baseline. Both LLM conditions open \$5–7 higher than the rule-based agent (\$65–67 vs. \$60). The rule-based buyer opens at exactly 50% of its value ceiling, while the LLM agents do not reproduce this heuristic precisely.

Finding 2: Chain-of-thought prompting produces more conservative counter-offers. The deliberative agent's seller counter (\$96.57) is \$7.21 lower than the reactive agent's (\$103.78), yielding a concession gap of \$29.83 versus \$38.32. This is a prompting effect—the BELIEFS→TARGET→STRATEGY→ACTION chain leads the model to hold a firmer position—rather than evidence of Rubinstein-style time-preference patience. H-A3 is confirmed: structured reasoning alters counter-offer behaviour.

Finding 3: Rule-based agent produces zero variance. Standard deviation of \$0.00 across all 150 sessions per round confirms that the seeded RNG and linear concession formula are fully deterministic, validating the platform's reproducibility guarantees.

Limitation. The feasibility gate forces settlement at round 2 in all sessions, meaning only two data points per session are observed. A full concession trajectory over 10 rounds is not available in this experiment. Future work should test with the gate disabled or a narrower ZOPA to observe multi-round concession curves.

6.3 Experiment B: Anchoring Effects

Table 4 reports deal rates, first offers, and settlement prices across the three ZOPA conditions.

Table 4: Anchoring experiment outcomes by condition ($n = 150$ per condition, pooled across 3 seeds). Seller cost is fixed at \$80 in all conditions.

Condition	Deal rate	First offer	Final price	Buyer surplus
`anchor_low` (ZOPA = \$20)	2.7% (4/150)	\$55.00 ± \$0.00	\$96.40 ± \$0.00	\$3.60
`anchor_mid` (ZOPA = \$40)	100% (150/150)	\$65.51 ± \$1.03	\$103.48 ± \$3.25	\$16.52
`anchor_high` (ZOPA = \$70)	100% (150/150)	\$75.00 ± \$0.00	\$101.55 ± \$5.50	\$48.45

First-offer / final-price Pearson $r = -0.148$ across all deals.

Finding 4: ZOPA width is the primary determinant of deal success. The `anchor_low` condition collapses to a 2.7% deal rate compared to 100% in the other two conditions. With ZOPA = \$20, the seller's opening counter of approximately \$104 exceeds the buyer's hard cap of \$100, making the offer infeasible. The feasibility gate cannot fire, and the LLM must negotiate the seller below its natural opening position—a task it nearly always fails. H-B1 is confirmed.

Finding 5: Classical anchoring bias is absent (null result). Pearson $r = -0.148$ indicates no meaningful positive relationship between first offer and final price. The `anchor_high` condition (first offer \$75) does not produce higher settlement prices than `anchor_mid` (first offer \$65); final prices converge to the \$101–103 range regardless. H-B2 is rejected; H-B2′ is supported.

This is a departure from human bargaining results, where first offers reliably anchor final outcomes [5]. One possible explanation is that the accept-condition block in the prompt provides explicit surplus information, reducing the model's reliance on the first offer as an informational signal. This hypothesis has not been tested via ablation.

Design confound. The independent variable simultaneously manipulates ZOPA width and first-offer level, making it impossible to fully isolate anchoring from ZOPA feasibility effects. The Pearson r metric partially addresses this by measuring within-deal correlation, but the confound remains.

6.4 Experiment C: Deadline Pressure

Table 5 reports settlement timing across the three horizon conditions.

Table 5: Deadline experiment outcomes by time horizon ($n = 150$ per condition, pooled across 3 seeds).

Max rounds	Deal rate	Avg closing round	SD	Last-2-round share
4	100%	3.00	0.00	100.0%
8	100%	3.00	0.00	0.0%
16	100%	3.00	0.00	0.0%

Finding 6: Deadline horizon does not affect settlement timing (null result). All conditions close at exactly round 3 with zero variance, regardless of whether the time limit is 4, 8, or 16 rounds. The feasibility gate fires deterministically after the seller's round-1 counter-offer, producing a mechanical settlement at round 2 that is fully independent of deadline proximity. H-C1 is rejected; H-C1′ is confirmed.

Finding 7: The `max_rounds=4` last-2-round share is an arithmetic artifact. The 100% last-2-round share under `max_rounds=4` occurs because round 3 of 4 falls within the "last 2 rounds" window. This is not evidence of deadline-driven concession; it is a consequence of the gate's fixed firing time relative to a short horizon.

Confounds. Two design choices compound to produce this null result. First, the feasibility gate settles deals before deadline proximity becomes relevant. Second, the prompt steers agents to prefer COUNTER over REJECT, removing the primary mechanism by which agents could delay settlement. To study genuine deadline effects, future work should disable the gate and remove the prefer-counter instruction, accepting lower deal rates as part of the measurement.

6.5 Experiment D: Market Price Discovery

Tables 6 and 7 report aggregate market statistics and temporal comparison for the 30-tick market simulation.

Finding 8: Market prices stabilise rapidly. The price trend slope of +\$0.057/tick is economically negligible (total drift < \$2 over 30 ticks against a mean of \$89). The market stabilises within the first few ticks, consistent with the prediction that decentralised bilateral bargaining converges to a stable price level even without a central auctioneer [15]. H-D1 is confirmed.

Finding 9: Law of One Price fails; price dispersion persists. Within-tick standard deviation of \$20 means that simultaneous transactions occur at substantially different prices within the same market period. This is the expected outcome under bilateral bargaining, where each pair negotiates indepen-

Table 6: Market dynamics aggregate statistics (30 ticks, single seed).

Metric	Value
Mean transaction price	\$89.44
Price range	\$78.12 – \$110.03
Within-tick price std (mean)	\$20.02
Average liquidity	52.2%
Liquidity range	13.3% – 80.0%
Price trend slope	+\$0.057 / tick
Buyer surplus (mean)	\$32.95
Seller surplus (mean)	\$16.40

Table 7: Early vs. late market comparison (single seed).

Period	Mean price	Mean liquidity
Ticks 0–4 (early)	\$84.18	54.7%
Ticks 25–29 (late)	\$86.57	61.3%

dently and prices reflect the specific ZOPA of each pair rather than a single market-clearing price. H-D2 is confirmed.

Finding 10: Buyers capture more surplus than sellers. Mean buyer surplus (\$32.95) exceeds mean seller surplus (\$16.40) by a factor of two. The mean transaction price of \$89.44 lies below the ZOPA midpoint for most pairs, consistent with a first-mover advantage: buyers open in round 0, anchoring the negotiation toward the lower end of the ZOPA. H-D3 is confirmed.

Finding 11: Random matching is liquidity-inefficient. Average liquidity of 52.2% means that nearly half of all matched pairs fail to trade. With random matching, many pairs are matched across non-overlapping value/cost ranges (buyer value < seller cost), making trade impossible regardless of agent capability.

Single-seed caveat. These results are from a single seed (42). The specific agent pairings, ZOPA distributions, and tick-level outcomes may vary under different seeds. The findings are suggestive but should not be treated as confirmatory.

6.6 Experiment E: Shock Response

Table 8 compares market outcomes under the no-shock baseline and the shock condition.

Table 8: Shock response outcomes: `no_shock` vs. `with_shock` (30 ticks each, single seed).

Metric	`no_shock`	`with_shock`	Δ
Mean price	\$90.92	\$88.77	−\$2.15
Price min	\$78.07	\$69.95	−\$8.12
Price max	\$104.76	\$100.82	−\$3.94
Within-tick price std	\$20.36	\$19.55	−\$0.81
Mean liquidity	54.2%	51.1%	−3.1 pp
Liquidity min	26.7%	13.3%	−13.4 pp
Liquidity max	86.7%	86.7%	0

Finding 12: Shocks reduce prices and liquidity. The shock condition produces a lower mean price (\$88.77 vs. \$90.92) and lower average liquidity (51.1% vs. 54.2%). Random demand and supply multipliers in $[0.7, 1.3]$ with 30% probability compress the effective ZOPA for affected pairs, reducing both the number of successful trades and the price level at which they occur. H-E2 is confirmed.

Finding 13: Shocks produce asymmetric liquidity tail risk. The worst-case tick liquidity drops sharply with shocks (13.3% vs. 26.7%), while the best-case tick remains unchanged (86.7% in both conditions). Negative shocks that simultaneously suppress buyer values and inflate seller costs can eliminate nearly all viable ZOPA pairs in a single tick. Positive shocks widen the ZOPA but cannot push liquidity above the random-matching ceiling. H-E3 is confirmed.

Finding 14: Price dispersion is unchanged by shocks. Within-tick price standard deviation is nearly identical across conditions (\$20.36 vs. \$19.55). The bilateral bargaining protocol—in which each pair negotiates independently—is the primary driver of price dispersion, not the width of the ZOPA. The shock mechanism shifts the ZOPA but does not change the negotiation process. H-E1 is partially rejected: while shocks affect price levels and liquidity, they do not increase within-tick price dispersion.

Single-seed caveat. As with Experiment D, these results are from a single seed. The specific shock realisations (which ticks are shocked, by how much) are fixed; different seeds would produce different shock sequences and potentially different aggregate effects.

6.7 Cross-Experiment Analysis

Tables 9 and 10 synthesise the empirical patterns observed across all five experiments.

Table 9: Where LLM agents align with economic theory.

Economic principle	Evidence	Exp
ZOPA determines deal success	2.7% vs. 100% deal rate as ZOPA narrows	B
Market price convergence	+$0.057/tick slope; stable $89 equilibrium	D
Bilateral bargaining dispersion	Within-tick σ = $20; Law of One Price fails	D
First-mover buyer advantage	Buyer surplus $32.95 > seller surplus $16.40	D
Shock → price/liquidity reduction	−$2.15 price, −3.1 pp liquidity	E
Asymmetric shock tail risk	Worst-case liquidity drops 13.4 pp; best-case unchanged	E

Table 10: Where LLM agents depart from classical predictions.

Predicted behaviour	Observed	Interpretation	Exp
Anchoring bias	$r = -0.148$	Prompt provides explicit surplus info	B
Deadline concession	Closes at round 3 always	Gate + prefer-counter steering	C
Shocks increase dispersion	σ unchanged	Protocol drives dispersion, not ZOPA	E

Feasibility gate as cross-cutting confound. The gate drives approximately 40% of all acceptances. In fixed-scenario experiments (A–C), it fires at round 2 in every session, limiting concession data to two observations and rendering settlement timing invariant to deadline manipulation. In market experiments (D–E), gate involvement varies by session depending on the drawn ZOPA. The gate's contribution to deal rates and settlement timing cannot be separated from agent capability without ablation studies that disable the gate entirely.

First-mover advantage as protocol effect. Buyer surplus exceeds seller surplus in both the fixed-scenario concession experiment (A) and the market experiment (D). This pattern is consistent with the buyer-first turn order in the alternating-offer protocol: the buyer's opening offer anchors the negotiation toward the lower end of the ZOPA. This is a protocol effect, not an agent-type effect, and would persist regardless of the LLM used.

Prompt steering as uncontrolled variable. The prefer-counter instruction and the accept-condition block are active in all LLM conditions across all experiments. Their combined effect is to reduce rejection, accelerate settlement, and possibly suppress anchoring. Because neither has been ablated, they function as uncontrolled confounds rather than experimental variables. Separating their effects from the model's intrinsic bargaining behaviour is an important direction for future work.

7 Discussion

This section interprets the experimental results, situates them relative to prior work, and identifies the limitations of the current study.

7.1 LLM Agents as Bounded Rational Actors

The five experiments reveal a consistent pattern: LLM agents align with economic theory on outcomes that are structurally determined by the negotiation protocol and private constraints, but fail to reproduce the cognitive heuristics that characterise human bargaining behaviour.

On the structural side, agents reliably produce ZOPA-determined deal success (Experiment B), stable market prices with persistent within-tick dispersion (Experiment D), first-mover surplus advantage (Experiments A and D), and directionally correct responses to supply/demand shocks (Experiment E). These outcomes depend primarily on the existence and width of the zone of possible agreement, the alternating-offer protocol, and the random matching mechanism—not on whether the agent reasons strategically. A trivially rational agent that accepts any positive-surplus offer would produce similar results in wide-ZOPA scenarios. The gate, which implements exactly this logic, drives approximately 40% of all acceptances.

On the heuristic side, agents fail to exhibit anchoring bias ($r = -0.148$, Experiment B) and deadline-driven concession acceleration (Experiment C). In human bargaining, these are well-documented departures from full rationality: humans anchor because they lack complete information about the opponent's constraints [5], and they concede near deadlines because of time pressure and loss aversion [13]. The LLM agents in our experiments do not replicate these patterns.

This suggests a qualitatively different form of bounded rationality. Human negotiators satisfice using heuristics because full optimisation is computationally intractable. LLM agents are bounded in a different way—their behaviour is conditioned on the prompt rather than on accumulated experience, and their decisions are shaped by the information explicitly provided to them (including pre-computed utility) rather than by cognitive shortcuts developed through repeated interaction. Whether this constitutes a form of rationality, a prompt artifact, or both, cannot be determined from the current experiments alone.

We note that this interpretation is tentative. The null results on anchoring and deadline sensitivity are each confounded by at least two design choices (Section 7.3), and we cannot rule out that different prompt architectures, larger models, or gate-disabled experiments would produce different results.

7.2 The Role of Prompt Architecture

The prompt is the LLM agent's entire decision architecture. Every piece of information included or excluded shapes the agent's behaviour, making prompt design a first-order experimental variable rather than a background condition.

Chain-of-thought and counter-offer conservatism. The clearest prompt-driven effect is the \$7.21 difference between the deliberative and reactive seller counter-offers (Experiment A). The deliberative prompt's BELIEFS→TARGET→STRATEGY→ACTION structure leads the model to hold a firmer position on its first counter. This is consistent with Abdelnabi et al. [1], who find that deliberation-style prompting improves agreement quality in multi-agent negotiation games. However, we cannot determine whether the model performs genuine multi-step strategic reasoning or whether the structured format simply constrains the distribution of output prices. The observation is that structured prompting produces a measurably different—and in economic terms, more conservative—opening position.

Accept-condition block and the anchoring null. The accept-condition block pre-computes the utility of the opponent's standing offer and presents it directly in the prompt. In human anchoring experiments [5], the anchoring effect depends on negotiators forming impressions under uncertainty about the opponent's reservation price. Our agents receive explicit surplus information, potentially overriding any tendency to anchor on the first offer. This is a plausible explanation for the anchoring null result, but it has not been tested: an ablation that removes the accept-condition block and forces the model to compute utility from raw constraints would be needed to confirm or reject this hypothesis.

Prefer-counter steering and the deadline null. The prompt instruction to prefer COUNTER over REJECT removes the primary mechanism by which agents could delay settlement. In combination with the feasibility gate (which settles deals before deadlines approach), this instruction makes deadline sensitivity structurally impossible to observe in the current design. The instruction was added for a practical reason: without it, small LLMs frequently reject profitable offers, producing meaningless 0% deal rates. This illustrates the tension between engineering reliability and experimental cleanliness that pervades LLM agent research.

Prompt as cognitive architecture. These observations point to a broader methodological concern. In human negotiation research, the cognitive architecture of the bargainer is relatively fixed; experimental variation comes from the task, the opponent, and the information structure. In LLM negotiation, the prompt *is* the cognitive architecture, and any change to the prompt—adding a reasoning scaffold, including or excluding surplus information, steering toward or away from rejection—can alter outcomes. This means that LLM negotiation results are inherently prompt-conditional. Generalising findings across prompt designs requires systematic ablation, which the current study does not provide.

7.3 Feasibility Gate: Necessity and Confound

The feasibility gate is the single most important design decision in the platform. It simultaneously enables and constrains the experimental programme.

Engineering necessity. Small language models (3B parameters) frequently fail to output ACCEPT even when the opponent's offer is clearly profitable, a failure mode documented in our early development iterations that motivated the gate's implementation. Without the gate, wide-ZOPA scenarios produce near-zero deal rates due to LLM output quality rather than strategic behaviour, rendering experimental comparison meaningless. The gate is not an experimental intervention but a practical prerequisite for conducting experiments with small models.

What the gate masks. In fixed-scenario experiments (A–C), the gate fires at round 2 in every session, limiting concession data to two observations per session and making settlement timing invariant to deadline manipulation. This is the primary reason Experiment C produces a null result: deals settle before deadline proximity becomes relevant. Extended concession trajectories are unobservable, and 100% deal rates in wide-ZOPA scenarios may reflect gate capability rather than agent capability.

What the gate reveals. The post-LLM override rate of 0% is itself informative: when the gate does not pre-empt the LLM call, the model never rejects a clearly feasible offer. This suggests that `llama3.2:3b` can recognise profitable offers when given explicit constraint information, even if it sometimes fails to generate ACCEPT autonomously.

In narrow-ZOPA scenarios (Experiment B, `anchor_low`), the gate cannot fire because the seller's opening counter exceeds the buyer's hard cap. Here, the LLM must negotiate on its own merits—and it fails 97.3% of the time. This is the most direct measure of the model's unassisted negotiation capability: it cannot bridge a $20 ZOPA when the opponent's opening position is outside the zone of agreement.

Recommendations. The gate should be treated as a controllable experimental variable in future work. Gate-on versus gate-off ablation studies would reveal the extent to which deal rates, concession trajectories, and settlement timing are agent-driven versus mechanically enforced. For experiments targeting deadline pressure, multi-round concession dynamics, or communication effects (Experiments F and G), the gate should be disabled, with the resulting lower deal rates accepted as part of the measurement rather than engineered away.

7.4 Comparison to Prior Work

We compare our findings to the prior work reviewed in Section 3.

Generative agent simulations. Park et al. [11] demonstrated that LLM agents exhibit emergent social behaviours in open-ended environments. Our market experiments (D and E) extend this paradigm to economic settings: LLM agents produce emergent market structure—stable price equilibrium, persistent price dispersion, and surplus asymmetry—without being explicitly programmed to produce these aggregate outcomes. The key difference is that our environment is constrained (hard budget and cost limits, structured protocol) rather than open-ended, enabling direct comparison against theoretical predictions.

LLM negotiation. Abdelnabi et al. [1] find that deliberation prompting improves agreement quality in multi-party, multi-issue negotiation games. Our Experiment A provides a complementary finding in a simpler setting: deliberative prompting produces more conservative counter-offers ($7.21 difference), which is consistent with improved strategic positioning though we observe it through a different metric (offer price rather than Pareto efficiency). We note that our setting is single-issue bilateral bargaining, which isolates price dynamics from the issue-trading effects present in their multi-issue design.

Mao et al. [9] show that personality-assigned prompts shift negotiation outcomes. Our prefer-counter instruction and communication strategies function similarly as prompt-level modifiers. The methodological difference is that we disclose prompt steering as a confound rather than studying it as a controlled variable, though the infrastructure for the communication-strategy experiment (Experiment G) exists for future work.

Lewis et al. [8] found that end-to-end trained negotiation agents develop non-human-like strategies, including degenerate repetitive behaviour. Our LLM agents show a different form of departure from human norms: rather than degenerating, they fail to reproduce specific human heuristics (anchoring, deadline pressure) while maintaining otherwise reasonable bargaining behaviour. The underlying mechanism differs—Lewis et al.'s agents are trained via reinforcement learning and self-play, while ours are prompted—but both findings suggest that artificial agents do not spontaneously replicate the full spectrum of human negotiation behaviour.

Classical theory. Our positive results confirm several qualitative predictions from bilateral bargaining theory. Market prices stabilise rapidly and exhibit persistent within-tick dispersion (consistent with Rubinstein and Wolinsky [15]), ZOPA width determines deal feasibility (consistent with Raiffa [12]), and the buyer-first protocol produces buyer-favourable surplus splits (consistent with first-mover advantage in alternating-offer models [14]). Shocks reduce prices and liquidity in directions consistent with supply and demand theory, though the magnitudes are modest and from a single-seed run.

Our null results depart from two well-established human findings. The absence of anchoring bias contradicts Galinsky and Mussweiler [5], though the departure is plausibly explained by the prompt's provision of explicit surplus information—a condition that does not hold in human experiments. The absence of deadline sensitivity contradicts Roth et al. [13], though this result is heavily confounded by the feasibility gate and prompt steering, and should not be interpreted as evidence that LLM agents lack the capacity for deadline awareness under different experimental conditions.

7.5 Limitations

We identify ten limitations of the current study, ordered by their impact on the interpretation of results.

1. **Single model.** All results are specific to `llama3.2:3b` at temperature 0.2. Larger models, different architectures, or fine-tuned models may exhibit anchoring bias, deadline sensitivity, or other behaviours absent in our experiments. No cross-model comparison is provided.

2. **Gate dominance.** In wide-ZOPA scenarios, the feasibility gate drives all acceptances. Deal rates and settlement timing in Experiments A–C reflect gate logic rather than agent decision-making. Without gate ablation, agent capability cannot be separated from gate enforcement.

3. **Two-round data.** The gate fires at round 2 in all fixed-scenario sessions, yielding only two data points per session for concession analysis. Multi-round concession dynamics—the core phenomenon the concession experiment was designed to study—are not observed.

4. **No prompt ablation.** The accept-condition block, prefer-counter steering, and deadline-salience language are all active but none are individually ablated. Their separate effects on anchoring, deal rates, and deadline sensitivity are unknown.

5. **Single seed for market experiments.** Experiments D and E use seed 42 only. The specific agent pairings, ZOPA distributions, and shock realisations are fixed. Results are suggestive but not confirmatory.

6. **Confounded anchoring design.** The independent variable in Experiment B simultaneously manipulates ZOPA width and first-offer level. The Pearson r metric partially addresses this, but anchoring cannot be fully isolated from ZOPA feasibility effects.

7. **No statistical tests.** All analyses are descriptive (means, standard deviations, Pearson r). No hypothesis tests, confidence intervals, or effect sizes are reported. The three-seed design in Experiments A–C provides limited replication; Experiments D–E have no replication.

8. **Single-price negotiation.** All sessions negotiate a single price. Real markets involve quality, delivery terms, and bundling. Multi-issue negotiation may produce qualitatively different dynamics, including the logrolling and issue-trading effects studied by Lewis et al. [8].

9. **No cross-run learning.** Agents do not accumulate experience across sessions within the completed experiments. The memory and reputation agents are implemented but untested. Whether LLM agents adapt strategies over repeated interactions remains an open question.

10. **LLM non-determinism.** Even at temperature 0.2, model outputs are not fully deterministic. The seeded RNG controls market-level randomness (population generation, matching, shocks) but not LLM inference randomness. This introduces uncontrolled variance into all LLM-agent conditions.

7.6 Implications

For automated negotiation deployment. LLM agents settle bilateral deals reliably when the ZOPA is wide, but fail at a 97.3% rate when the ZOPA narrows to $20 (Experiment B). Practical systems that deploy LLM negotiators should estimate ZOPA width before entering negotiation, and provide deterministic fallbacks—analogous to our feasibility gate—for scenarios where the model is likely to fail. The absence of anchoring bias may be advantageous in adversarial settings (the LLM will not be manipulated by aggressive first offers), though it also means the LLM will not exploit anchoring strategically.

For experimental methodology. The feasibility gate illustrates a general tension in LLM agent research: engineering interventions that make experiments viable can simultaneously mask the behaviours those experiments are designed to study. Researchers using LLM agents in structured settings should treat such interventions as controllable variables, report their involvement rates, and conduct ablation studies to separate agent capability from system-level enforcement.

Prompt architecture is a first-order experimental variable. Our results show that including or excluding specific information (accept-condition block, reasoning scaffold) produces measurable differences in bargaining behaviour. LLM negotiation results should be understood as prompt-conditional, and claims about LLM negotiation behaviour in general should be hedged accordingly.

For economic theory. The positive results suggest that certain market-level predictions of bilateral bargaining theory—price convergence, price dispersion, first-mover advantage, directional shock response—are robust to the specific cognitive architecture of the agents. These properties emerge from the structure of bilateral bargaining itself (independent pairwise negotiation, private constraints, alternating offers) rather than from agent sophistication. A 3B-parameter model produces qualitatively similar market structure to what theory predicts for rational agents, not because it is rational, but because the protocol and constraints do most of the work.

This is a modest finding. It does not demonstrate that LLM agents are economically rational, nor that they can substitute for human participants in experimental economics. It demonstrates that the structural predictions of bilateral bargaining theory hold under a relatively weak rationality assumption—that agents accept offers yielding positive surplus when explicitly informed of that surplus—which the gate enforces and the LLM, when queried, also satisfies.

7.7 Future Work

Several directions would strengthen and extend the current findings.

Gate and prompt ablation. The most immediate priority is to repeat Experiments A–C with the gate disabled and to separately ablate the accept-condition block, prefer-counter steering, and deadline-salience language. This would reveal which null results are gate artifacts and which reflect intrinsic LLM behaviour.

Multi-model comparison. Repeating key experiments with models of different sizes (7B, 13B, 70B) and architectures would test whether anchoring bias and deadline sensitivity emerge at larger scales. This is the single most important external-validity question.

Experiments F–H. The platform supports three additional experiment families—reputation and repeated interaction, communication strategy ablation, and matching mechanism comparison—for which infrastructure is implemented but empirical evaluation is deferred. These experiments should disable the gate to allow strategic adaptation and prompt-driven effects to manifest.

Multi-seed market experiments. Experiments D and E should be replicated across multiple seeds to assess whether the market-level findings (price convergence, surplus asymmetry, shock response) are robust or specific to the seed-42 random matching.

Statistical analysis. Formal hypothesis tests, confidence intervals, and effect-size measures would strengthen the evidential basis of the current findings. The three-seed design in Experiments A–C supports seed-level analysis of variance; the single-seed market experiments require additional replication before statistical testing is meaningful.

Human–LLM comparison. Running the same experimental conditions with human participants would enable direct comparison of bargaining behaviour. This would clarify whether the absence of anchoring and deadline effects is specific to LLMs or whether it extends to any agent operating with explicit constraint information.

8 Conclusion

This thesis presented a controlled experimental platform for evaluating LLM bargaining behaviour in alternating-offer bilateral negotiation under private economic constraints. The platform enforces hard budget and cost limits, provides a configurable deterministic settlement gate, logs every agent action at sufficient granularity to separate LLM-generated decisions from mechanically enforced settlements, and connects micro-level bargaining behaviour to macro-level market outcomes within a single system. Using this platform, we conducted five experiments totalling 2,250 negotiation sessions with a single 3B-parameter language model (`llama3.2:3b`).

The experiments produced two categories of findings. First, LLM agents reproduced several structural predictions of classical bargaining and market theory: deal success was determined by the width of the zone of possible agreement, market prices stabilised rapidly with persistent within-tick dispersion, the buyer-first protocol produced systematic surplus asymmetry, and exogenous shocks shifted prices and liquidity in theoretically predicted directions. These outcomes arise primarily from the negotiation protocol and private constraints rather than from agent sophistication—a 3B-parameter model, assisted by a feasibility gate that drives approximately 40% of acceptances, is sufficient to produce qualitatively correct market structure.

Second, agents failed to exhibit two well-documented human bargaining heuristics: anchoring bias and deadline-driven concession acceleration. These null results are informative but heavily confounded. The anchoring null may reflect the prompt's provision of explicit surplus information, which removes the uncertainty that drives anchoring in human negotiators. The deadline null is confounded by the feasibility gate (which settles deals before deadlines approach) and by prompt instructions that discourage rejection. Neither confound has been ablated; the null results characterise LLM behaviour under these specific experimental conditions, not LLM negotiation capability in general.

A recurring methodological theme is the tension between engineering reliability and experimental cleanliness. The feasibility gate was necessary to make small-model experiments viable—without it, wide-ZOPA scenarios produce near-zero deal rates due to LLM output quality—but it simultaneously masks the multi-round dynamics that several experiments were designed to study. More broadly, the prompt functions as the agent's entire cognitive architecture, making all results inherently prompt-conditional. These observations apply beyond our specific platform: researchers using LLM agents in structured economic settings should treat engineering interventions and prompt design choices as first-order experimental variables, disclose their involvement rates, and conduct ablation studies to separate agent capability from system-level enforcement.

Three directions for future work are most pressing. First, gate and prompt ablation—disabling the gate and separately removing the accept-condition block, prefer-counter steering, and deadline-salience language—would reveal which null results are artifacts and which reflect intrinsic model behaviour. Second, repeating key experiments with models of different sizes and architectures would test whether anchoring and deadline effects emerge at larger scales, addressing the most important external-validity question. Third, the platform supports three additional experiment families—reputation and repeated interaction, communication strategy ablation, and matching mechanism comparison—for which infrastructure is implemented but empirical evaluation is deferred to future work.

The platform is a testbed, not a finished product. Its value lies in enabling systematic, reproducible comparison of LLM bargaining behaviour against well-characterised theoretical predictions—including honest reporting of where that behaviour aligns with theory and where it does not.

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